

XIANYU CHEN

Kenneth H. Keller Hall 6-240, 200 Union St SE, Minneapolis, MN, 55455
(612)-814-3793 $\diamond$ chen6582@umn.edu $\diamond$ [chenxy99.github.io](https://github.com/chenxy99) $\diamond$ [Google Scholar](#)

EDUCATION

University of Minnesota, Twin Cities (GPA: 4.0/4.0) Ph.D. in Computer Science Advisor: Dr. Catherine Qi Zhao	Sept. 2019 - Dec. 2025
Sun Yat-sen University M.E. in Information and Communication Engineering Advisor: Dr. Ming Jiang	Sept. 2015 - June 2018
Sun Yat-sen University B.E. in Communication Engineering Advisor: Dr. Ming Jiang	Sept. 2011 - June 2015

RESEARCH INTEREST

Computer Vision, Machine Learning, Deep Learning, Nature Language Processing

PUBLICATION

Journals

- [J.7] **Xianyu Chen**, Jinhui Yang, Shi Chen, Louis Wang, Ming Jiang, and Qi Zhao. [Every Problem, Every Step, All In Focus: Learning to Solve Real-World Problems with Integrated Attention](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, Feb 2024.
- [J.6] **Xianyu Chen**, Yali Wang, Jianzhuang Liu and Yu Qiao. [DID: Disentangling-Imprinting-Distilling for Continuous Low-Shot Detection](#). *IEEE Transactions on Image Processing (TIP)*, vol. 29, pp. 7765-7778, July 2020.
- [J.5] **Xianyu Chen** and Ming Jiang. [Enhanced Adaptive Polar-Linear Interpolation Aided Channel Estimation](#). *IEEE Wireless Communications Letters (WCL)*, vol. 8, no. 3, pp. 693-696, June 2019.
- [J.4] Huaiyin Lu, Lin Zhang, **Xianyu Chen** and Zhiqiang Wu. [Recursive Carrier Interferometry Aided High Data Rate OFDM Systems with PAPR Suppression, Phase Noise Rejection and Carrier Frequency Offsets Compensation](#). *IEEE Transactions on Vehicular Technology (TVT)*, vol. 68, no. 4, pp. 3655 - 3671, April 2019.
- [J.3] Kuan Wu, Ming Jiang, Fengxia She and **Xianyu Chen**. [Relay-Aided Request-Aware Distributed Packet Caching for Device-to-Device Communication](#). *IEEE Wireless Communications Letters (WCL)*, vol. 8, no. 2, pp. 217-220, Feb. 2019.
- [J.2] Zhengpeng Li, Ming Jiang, Xiaona Zhang, **Xianyu Chen** and Weikun Hou. [Space-Time-Multiplexed Multi-Image Visible Light Positioning System Exploiting Pseudo-Miller-Coding for Smart Phones](#). *IEEE Transactions on Wireless Communications (TWC)*, vol. 16, no. 12, pp. 8261-8274, Dec. 2017.
- [J.1] **Xianyu Chen** and Ming Jiang. [Adaptive Statistical Bayesian MMSE Channel Estimation for Visible Light Communication](#). *IEEE Transactions on Signal Processing (TSP)*, vol. 65, no. 5, pp. 1287-1299, March 2017.

Conferences

- [C.11] **Xianyu Chen**, Ming Jiang and Qi Zhao. [GazeXplain: Learning to Predict Natural Language Explanations of Visual Scanpaths](#). In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024. (**Oral Paper, 2.3% acceptance rate**)

- [C.10] **Xianyu Chen**, Ming Jiang and Qi Zhao. [Beyond Average: Individualized Visual Scanpath Prediction](#). In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [C.9] Jinhui Yang*, **Xianyu Chen***, Ming Jiang, Shi Chen, Louis Wang and Qi Zhao. [VisualHow: Multimodal Problem Solving](#). In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. (*Co-first authors/Equal contribution)
- [C.8] **Xianyu Chen**, Ming Jiang and Qi Zhao. [Leveraging Human Attention in Novel Object Captioning](#). In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
- [C.7] **Xianyu Chen**, Ming Jiang and Qi Zhao. [Predicting Human Scanpaths in Visual Question Answering](#). In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [C.6] **Xianyu Chen**, Ming Jiang and Qi Zhao. [Self-Distillation for Few-Shot Image Captioning](#). In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021.
- [C.5] Ze Yang, Yali Wang, **Xianyu Chen**, Jianzhuang Liu and Yu Qiao. [Context-Transformer: Tackling Object Confusion for Few-Shot Detection](#). In *AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
- [C.4] **Xianyu Chen** and Ming Jiang. [Low-Complexity Adaptive Channel Estimation](#). In *Proceedings of the Vehicular Technology Conference (VTC)*, 2018.
- [C.3] Yufa Chen, Ming Jiang, Lin Zhang and **Xianyu Chen**. [Polarity Modulated Complex Colour Shift Keying for OFDM-Based Visible Light Communication](#). In *Proceedings of the IEEE/CIC International Conference on Communications in China (ICCC)*, 2017.
- [C.2] Zhengpeng Li, Ming Jiang, Xiaona Zhang, **Xianyu Chen** and Weikun Hou. [Miller-Coded Asynchronous Visible Light Positioning System for Smart Phones](#). In *Proceedings of the IEEE Vehicular Technology Conference (VTC)*, 2017.
- [C.1] **Xianyu Chen** and Ming Jiang. [Enhanced Bayesian MMSE Channel Estimation for Visible Light Communication](#). In *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2016.

Preprint

- [R.1] **Xianyu Chen**, Ming Jiang and Qi Zhao. [Leveraging Bottom-Up and Top-Down Attention for Few-Shot Object Detection](#). In *arXiv*, 2020.

Patent

- [P.14] Qi Zhao, Ming Jiang, **Xianyu Chen** and Zhi Yang. [Method, apparatus and system for recognizing tremor symptom, recognition terminal and storage medium](#), 2023, U.S. Patent Application No. 18/448,995.
- [P.13] Qi Zhao, Ming Jiang, **Xianyu Chen** and Zhi Yang. [Tremor symptom identification method, device, identification terminal, storage medium and system](#), 2024, China Patent No. CN118196594A.
- [P.12] Ming Jiang and **Xianyu Chen**, [Dimming control visible light communication system and method based on convex optimization framework](#), 2019, China Patent No. CN110855364B (Granted).
- [P.11] Ming Jiang and **Xianyu Chen**, [Optimal dimming control visible light communication system and method based on distortion strategy](#), 2019, China Patent No. CN110855363B (Granted).
- [P.10] Ming Jiang, Zhengpeng Li, Kunyi Cai and **Xianyu Chen**, [Dual-mode LED positioning method and system based on different Lambertian radiation lobe moduli](#), 2018, China Patent No. CN109324311B (Granted).
- [P.9] Yu Qiao, Ze Yang, Yali Wang, **Xianyu Chen**, Jianzhuang Liu and Jun Yue, [Image processing method and related equipment](#), 2020, China Patent Application No. CN113221929A
- [P.8] Yu Qiao, **Xianyu Chen** and Yali Wang, [Target detection method, device and equipment for continuous small sample images](#), 2019, China Patent No. CN110033026B (Granted).

- [P.7] Ming Jiang, Junyu Chen, Lei Zhao and **Xianyu Chen**, [Channel estimation method of marine communication system](#), 2020, China Patent No. CN111404847B (Granted)
- [P.6] Ming Jiang, Zefeng Lin and **Xianyu Chen**, [OFDM system supporting color adjustment and CSK constellation diagram detection](#), 2017, China Patent No. CN108092714B (Granted).
- [P.5] Ming Jiang, Yufa Chen, Lin Zhang and **Xianyu Chen**, [Optical OFDM communication system based on polar modulation and complex color shift keying](#), 2017, China Patent No. CN107395278B (Granted).
- [P.4] Ming Jiang, Kuan Wu, Fengxia She and **Xianyu Chen**, [Relay D2D data packet caching method based on user grouping demand diversity](#), 2018, China Patent No. CN108668261B (Granted).
- [P.3] Ming Jiang and **Xianyu Chen**, [Channel estimation method based on enhanced self-adaptive polarization linear interpolation](#), 2017, China Patent No. CN107508777B (Granted).
- [P.2] Ming Jiang, Jianhui Li, **Xianyu Chen**, Yixue Lei and Yunfei Zhang, [Internet-of-vehicles human-vehicle resource allocation method based on geographic region information](#), 2017, China Patent No. CN107659915B (Granted).
- [P.1] **Xianyu Chen** and Ming Jiang, [Visible light channel estimation method and system](#), 2015, China Patent No. CN105471777B (Granted).

TALK

- “GazeXplain: Learning to Predict Natural Language Explanations of Visual Scanpaths”, presented at *European Conference on Computer Vision (ECCV)*, 2024.
- “Leveraging Human Attention in Novel Object Captioning”, presented at *International Joint Conference on Artificial Intelligence (IJCAI)*, 2021.
- “Self-Distillation for Few-Shot Image Captioning”, presented at *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2021.

SERVICE

Review for Journals

- | | |
|---|----------|
| • IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) | 2023-Now |
| • IEEE Transactions on Multimedia (TMM) | 2023-Now |
| • IEEE Transactions on Neural Networks and Learning Systems (TNNLS) | 2023-Now |
| • IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) | 2025-Now |
| • ACM Transactions on Multimedia Computing Communications and Applications (TOMM) | 2025-Now |
| • Journal of Visual Communication and Image Representation | 2021-Now |

Review for Conferences

- | | |
|---|------------------------------|
| • International Conference on Computer Vision (ICCV) | 2023, 2025 |
| • European Conference on Computer Vision (ECCV) | 2022, 2024, 2026 |
| • Computer Vision and Pattern Recognition (CVPR) | 2022, 2023, 2025, 2026 |
| • Winter Conference on Applications of Computer Vision (WACV) | 2022, 2023, 2024, 2025, 2026 |
| • AAAI Conference on Artificial Intelligence (AAAI) | 2023, 2024, 2025, 2026 |

ACADEMIC EXPERIENCE

Visual Information Processing Lab, University of Minnesota

Sept. 2019 - Dec. 2025

Computer Vision, Vision and Language, Machine Learning

- Proposed a GazeXplain, a novel study of visual scanpath prediction and explanation.
- Proposed an individualized scanpath prediction (ISP), a new attention modeling task that aims to accurately predict how different individuals shift their attention in diverse visual tasks.
- Proposed a graph-based approach to vision-language problem solving for understanding and solving real-life problems.
- Proposed a large-scale dataset enabling a family of new vision-language tasks and computational methods for understanding and solving real-life problems.
- Designed a new method to complement novel object captioners with human attention features characterizing generally important information independent of tasks.
- Designed a new deep reinforcement learning method to predict scanpaths leading to different performances in visual question answering.
- Developed an ensemble-based self-distillation method for image captioning with few paired data and a large number of unpaired images and captions.
- Designed a new method for few-shot object detection by leveraging bottom-up and top-down attention based on object-concentration loss and background-concentration loss.

Multimedia Research Center, Shenzhen Institutes of Advanced Technology Chinese Academy of Sciences

Aug. 2018 - Dec. 2018

Computer Vision in Low-Shot Object Detection

- Designed a simple but effective solution for continuous low-shot detection based on architecture design (Disentangling), model initialization (Imprinting), and training methodology (Distilling).

Wireless Communication Lab, Sun Yat-sen University

Dec. 2013 - June 2018

Estimation Theory in Optical Wireless Communication

- Designed a dimming control framework achieving optimal waveform design and extended it into two different variants to acquire better performance.
- Designed an enhanced adaptive polar linear interpolation to estimate the channel information more robustly and efficiently.
- Designed a new channel estimation method and analyze its performance bound based on Bayesian MMSE estimation.

INDUSTRIAL EXPERIENCE

Fasikl Incorporated

Oct. 2025 - Now

Full-time Employee

- Research Scientist - AI. Research and Development Department.

Fasikl Incorporated

Jan. 2025 - Sept. 2025

Part-time Senior Intern

- Research Scientist. Research and Development Department.

Fasikl Incorporated <i>Part-time Intern</i>	Aug. 2024 - Dec. 2024
Research Scientist. Research and Development Department.	
Fasikl Incorporated <i>Intern</i>	May. 2024 - Aug. 2024
Research Scientist. Research and Development Department.	
Fasikl Incorporated <i>Intern</i>	May. 2023 - Aug. 2023
Research Engineer. Mentor: Linh Hoang.	

TEACHING EXPERIENCE

Teaching Assistant	
CSCI 5521: Machine Learning Fundamentals	Spring 2024
CSCI 5521: Machine Learning Fundamentals	Fall 2023
CSCI 2033: Linear Algebra	Spring 2023
CSCI 5521: Machine Learning Fundamentals	Fall 2022
CSCI 5302: Analysis of Numerical Algorithms	Spring 2022
CSCI 5521: Machine Learning Fundamentals	Fall 2021

RESEARCH EXPERIENCE

Research Assistant	
Visual Information Processing Lab	Summer 2022
Visual Information Processing Lab	Summer 2021
Visual Information Processing Lab	Summer 2020

HONOR

Distinguished Dissertation Award, Mathematics, Physical Sciences & Engineering	University of Minnesota, 2026
Nominated for CGS/ProQuest Distinguished Dissertation Award	University of Minnesota, 2026
Outstanding Reviewers	Computer Vision and Pattern Recognition (CVPR), 2026
Outstanding Reviewers	Winter Conference on Applications of Computer Vision (WACV), 2025
College Science and Engineering Graduate Fellowships	University of Minnesota, 2019, 2020
National Scholarship	Department of Education of China, 2013, 2017
The 1st Prize of Outstanding Students Scholarships	Sun Yat-sen University, 2013, 2014, 2015, 2016, 2017
Outstanding Undergraduate Student Award	Sun Yat-sen University, 2015

- Zhentai Scholarship

Sun Yat-sen University, 2012

AWARD

- The 2nd Prize in Contemporary Undergraduate Mathematical Contest in Modeling 2014
- Meritorious Winner in Mathematical Contest in Modeling 2013
- The 1st Prize in Chinese Mathematics Competitions 2013
- The 2nd Prize in National Undergraduate Electronics Design Contest 2013
- The 3rd Prize in Summer Camp of Undergraduate Mathematical Contest in Modeling 2012

SKILL

Programming Language: Python, Matlab, JavaScript, HTML, C/C++, Linux shell

Tools: Pytorch, Tensorflow, Keras, Opencv, Unix/Linux, Git, Scikit-Learn, LaTeX